CASE STUDY REPORT

Predictive Hardware Failure Analytics: Transforming Raw Data into Actionable Insights through Data Preprocessing, Analysis, and Visualization

A Data Science Case Study on Predictive Maintenance for Enterprise Hard Disk Drives, based on the Backblaze Hard Drive Data Set (Q1 2026)

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***Abstract*—**Unplanned hard-disk-drive failure is one of the most common causes of data loss and service downtime in data centres. This case study applies a complete data-science pipeline — preprocessing, exploratory data analysis (EDA), and visualization — to a large-scale, real-world hard-drive telemetry dataset in order to understand the statistical and behavioural signature of drive failure. The working dataset, drawn from Backblaze's publicly published daily hard-drive "Drive Stats" telemetry, contains 30,943,146 daily SMART records for 351,095 individual drives spanning 80 drive models over a 90-day observation window (1 January – 31 March 2026). Following a revision of the project's code, this edition reflects a change in how the three analytical stages are chained together: preprocessing (Member 2) still removes 345,662 exact duplicate records, imputes missing SMART readings by median/mode substitution, caps outliers using the IQR method, and standardizes 15 continuous features, producing a clean file of 30,597,484 records with zero missing values and zero duplicates. However, the EDA (Member 3) and visualization (Member 4) notebooks were both re-pointed to the original, uncleaned source file. As a result, every statistic and chart in Sections V and VI characterises the raw 30,943,146-row feed rather than the cleaned dataset — recovering real signal the earlier cleaned-and-imputed analysis had erased: SMART 188 (Command Timeout, raw) now shows the strongest measured association with failure of any feature ($r = +0.118$). The revised analysis confirms an extremely imbalanced target variable (only 1,054 of 30,943,146 records, 0.0034%, are failures) and identifies capacity_bytes and smart_9_raw (power-on hours) as strongly correlated ($r \approx -0.72$). Eight visualizations further characterise the distribution of drive capacities, failure proportions, and inter-feature relationships. Two methodological limitations introduced by the code revision — a non-random 500,000-row sampling slice and a floating-point negative-variance artifact — are documented as findings and carried into concrete recommendations for the next revision of the codebase and the next phase of model development.

Index Terms*—*predictive maintenance, SMART telemetry, data preprocessing, exploratory data analysis, correlation analysis, data visualization, class imbalance, hard disk drive failure

I. INTRODUCTION

Modern organisations increasingly rely on large fleets of hard disk drives (HDDs) to store business-critical data. Because drives fail without warning, unplanned failures create risk of data loss, degrade service availability, and raise operational costs associated with emergency replacement and RAID rebuilds. Predictive maintenance — using a drive's own internal health telemetry to anticipate failure before it happens — is widely regarded as a more efficient alternative to purely reactive or purely time-based maintenance strategies.

Every modern hard drive continuously records SMART (Self-Monitoring, Analysis and Reporting Technology) attributes: internal counters and sensor readings such as reallocated sector count, power-on hours, seek error rate, and temperature. Backblaze, a cloud storage and backup provider, has published daily SMART snapshots for its production drive fleet since 2013, creating one of the largest publicly available datasets for HDD-failure research. This case study uses a 2026 first-quarter extract of that dataset as the foundation for a complete, five-stage group data-science exercise: data collection, preprocessing, exploratory data analysis, visualization, and insight synthesis.

This report documents the full pipeline followed by the project team and consolidates the results produced at each stage into a single, decision-oriented narrative. It has been revised to reflect changes the team made to the preprocessing, EDA, and visualization notebooks since the previous edition — in particular, a change to which file the EDA and visualization stages read from — and calls out, explicitly, where that change altered the numbers and conclusions reported previously.

II. PROBLEM STATEMENT

Data-centre operators need to know, from daily SMART telemetry, which drives are at elevated risk of failing so that pre-emptive replacement or data migration can be scheduled. The problem this case study addresses is therefore twofold:

- **Data-quality problem:** the raw daily telemetry feed contains missing SMART values (ranging from under 1% to over 86% missing depending on the attribute), duplicate daily records, and extreme outliers, all of which must be resolved before the data can support reliable analysis or modelling.
- **Analytical problem:** drive failure is a rare event (a severe class-imbalance problem), so the study must characterise which SMART attributes, drive models, and capacity classes are statistically associated with failure, using descriptive statistics, correlation analysis, and visualization, as a foundation for future predictive modelling.

The scope of this case study is limited to preprocessing, exploratory analysis, and visualization of the telemetry feed; construction and evaluation of a trained failure-prediction classifier is identified as future work in Section VIII. A third, process-level problem emerged from this revision and is treated as an in-scope finding: keeping the preprocessing, EDA, and visualization notebooks reading from a single, agreed source file as the pipeline evolves.

III. DATASET DESCRIPTION

A. Source

The dataset used in this study is a daily hard-drive SMART telemetry extract in the style of Backblaze's publicly released "Hard Drive Data and Stats" (Drive Stats) dataset, published quarterly since 2013 for research and industry use. The working file, backblaze_analytical_dataset.csv, follows the standard Backblaze Drive Stats schema (date, serial_number, model, capacity_bytes, failure, and paired *_normalized / *_raw SMART attribute columns).

TABLE I. DATASET STRUCTURE AND CHARACTERISTICS

Characteristic	Value
Original records (rows)	30,943,146
Original attributes (columns)	19
Observation window	90 days (Jan–Mar 2026)
Unique drives	351,095
Distinct drive models	80
Target variable	failure (0/1)
Records after cleaning	30,597,484
Records used in this EDA	30,943,146 (raw file)

B. SMART Attributes Used

TABLE II. KEY SMART ATTRIBUTES

SMART ID	Attribute	Meaning
SMART 5	Reallocated Sectors	Sectors remapped after defect
SMART 9	Power-On Hours	Cumulative powered-on hours
SMART 90	Vendor-specific	Present on a subset of models
SMART 187	Reported Uncorrectable Errors	Errors not recovered by ECC
SMART 188	Command Timeout	Aborted ops due to timeout
SMART 197	Current Pending Sector	Unstable sectors awaiting remap
SMART 198	Uncorrectable Sector Count	Sectors failing ECC/verification

C. Fitness for the Case Study

With over 30 million records, 351,095 unique physical devices, a well-defined binary outcome (failure), and 14 continuous predictor variables, the dataset comfortably exceeds the assignment's minimum requirement of 500 records with multiple attributes. Its scale additionally required memory-aware engineering practices (chunked reading, dtype conversion, sparse encoding, float32 downcasting) throughout preprocessing, EDA, and visualization.

IV. DATA PREPROCESSING METHODOLOGY

Preprocessing was carried out in Python (pandas, NumPy, scikit-learn) on the full 30.9-million-row raw file, using memory-safe techniques so the pipeline could run on a standard workstation. This stage (Member 2) was re-run for this edition and produced results identical to the previous edition — the 21-step methodology is unchanged.

A. Missing-Value Analysis and Treatment

TABLE III. MISSING-VALUE TREATMENT

Column	Missing (%)	Treatment
smart_90_*	86.35%	Median imputation
smart_188_*	66.91%	Median imputation
smart_187_*	66.82%	Median imputation
smart_197_*	2.51%	Median imputation
smart_198_*	0.77%	Median imputation
smart_5_*	0.67%	Median imputation
smart_9_*	0.13%	Median imputation

All numeric columns were imputed using their column median (robust to skew); the three categorical columns had no missing values. After treatment, total missing values were reduced from 122,124,538 cell-level nulls to zero.

B. Duplicate Records

An exact row-level duplicate check identified 345,662 fully duplicated records (1.12% of the raw file), removed using drop_duplicates(). A second check after cleaning confirmed zero duplicates remained.

C. Outlier Detection and Treatment

Outliers were flagged using the 1.5×IQR rule and capped (winsorized) rather than deleted, preserving sample size while limiting the influence of extreme values.

TABLE IV. OUTLIERS DETECTED (1.5×IQR)

Feature	Outliers
smart_90_raw	3,941,911
capacity_bytes	1,868,933
smart_5_raw	1,570,312
smart_188_raw	662,232
smart_197_raw	612,655
smart_187_raw	457,208
smart_198_raw	423,526
smart_9_raw	4,462

D. Categorical Encoding

serial_number was retained (351,095 unique values, 1.15% of rows) since its uniqueness ratio fell below the 50% removal threshold. date and model were left in native form, exceeding the 20-category one-hot-encoding threshold set for memory safety; no categorical column required encoding in this pass.

E. Transformation and Standardization

No feature met the log1p() skew threshold (>1) after outlier capping. The 15 continuous SMART/capacity features were standardized with scikit-learn's StandardScaler, applied in float32 to control memory footprint — reducing the in-memory footprint from 9,190.94 MB (raw) to 7,337.45 MB (cleaned).

F. Preprocessing Summary

TABLE V. PREPROCESSING SUMMARY

Metric	Value
Original rows / columns	30,943,146 / 19
Duplicates removed	345,662
Remaining missing values	0
Categorical columns encoded	0
Columns standardized	15
Final rows / columns	30,597,484 / 19

The cleaned, standardized dataset (cleaned_disk_smart_data.csv) was exported by Member 2 as the intended single source of truth. As documented in Finding 6 (Section VII), the EDA and visualization notebooks in this revision were not re-pointed to this file, and instead read the raw source file directly.

V. EXPLORATORY DATA ANALYSIS

In this revision, the EDA notebook (Member 3) reads directly from the raw 30,943,146-row source file using a chunked, memory-efficient methodology, rather than the Member-2 cleaned file used previously. The statistics below describe the fleet's telemetry before deduplication, imputation, outlier capping, or standardization.

A. Descriptive Statistics (N = 30,943,146)

TABLE VI. DESCRIPTIVE STATISTICS (RAW FEED)

Feature	Mean	Median	Skew
capacity_bytes	1.565×10^{13}	1.600×10^{13}	-0.27
failure (0/1)	0.0000341	0	408.25
smart_9_normalized	63.15	76.0	-0.57
smart_9_raw (hrs)	31,146.5	28,002	0.39
smart_187_normalized	99.01	100.0	-10.79
smart_188_normalized	100.00	100.0	-335.03
smart_197_raw	4.72	0.0	576.30
smart_198_raw	3.43	0.0	591.31

Member 3's incremental variance formula produced a mathematically impossible negative variance for capacity_bytes (a floating-point cancellation artifact — see Finding 8, Section VII), rather than a property of the data itself.

B. Class Balance of the Target Variable

TABLE VII. CLASS BALANCE

failure	Record Count	Percentage
0 (healthy)	30,942,092	99.9966%
1 (failed)	1,054	0.0034%

This is an extreme class imbalance — roughly 1 failure event for every 29,357 healthy daily records — consistent with published Backblaze annualised failure rates well under 2%. This remains the single most important characteristic to account for in any future predictive model built on this data.

C. Correlation Structure

A Pearson correlation matrix was computed from a fixed, non-random sample of the first 500,000 rows of the file (a methodology limitation discussed as Finding 7). With that caveat, the strongest inter-feature relationships were:

TABLE VIII. STRONGEST CORRELATIONS

Feature 1	Feature 2	r
smart_197_raw	smart_198_raw	+0.994
smart_188_norm.	smart_197_raw	-0.932
smart_188_norm.	smart_198_raw	-0.932
smart_5_norm.	smart_197_norm.	+0.877
capacity_bytes	smart_9_raw	-0.719

TABLE IX. CORRELATION WITH FAILURE

Feature	r with failure
smart_188_raw	+0.1178 (strongest)
smart_187_normalized	-0.0367
smart_187_raw	+0.0054
smart_9_raw	+0.0019
capacity_bytes	-0.0012

This is the single most consequential change from the previous edition: smart_188_raw (Command Timeout) now shows a materially larger association with failure than either power-on hours or capacity — previously considered the dataset's only usable predictors. Because this is drawn from a small, non-random opening slice, it should be treated as a lead to validate against the full population, not a confirmed effect size.

D. Failure-Conditioned Comparison

Member 4's visualization notebook independently computed capacity_bytes by failure status across the full population: healthy mean 1.565×10^{13} bytes vs. failed mean 1.476×10^{13} bytes. Failed drives have, on average, smaller capacity than healthy drives — consistent with the previous edition and treated as authoritative here. Member 3's own group-mean calculation, based on only the first 50,000 rows, produced the opposite direction — a sign disagreement caused entirely by a small, non-random sample containing very few of the fleet's 1,054 failure events (Finding 7).

E. Anomaly Detection (|z| > 3)

TABLE X. Z-SCORE ANOMALIES (FIRST 500,000 ROWS)

Feature	Anomalies
capacity_bytes	5,510
smart_90_raw	1,603
smart_187_normalized	1,174
smart_5_raw	862
smart_188_raw	384
failure	3

F. Drive-Model Frequency

The fleet spans 80 distinct drive models. On the raw, pre-deduplication file, the five most common by daily-record volume — WDC WUH722222ALE6L4, TOSHIBA MG08ACA16TA, TOSHIBA MG07ACA14TA, ST16000NM001G, and WDC WUH721816ALE6L4 — together account for roughly 53% of all daily records, meaning fleet-level statistics are heavily influenced by the reliability of a small number of high-volume models.

VI. DATA VISUALIZATIONS

Eight charts were produced by Member 4. In this revision, all eight were generated from the raw 30,943,146-row source file rather than the cleaned/standardized dataset, so axes are shown in native units and every correlation and count reflects the pre-cleaning feed.

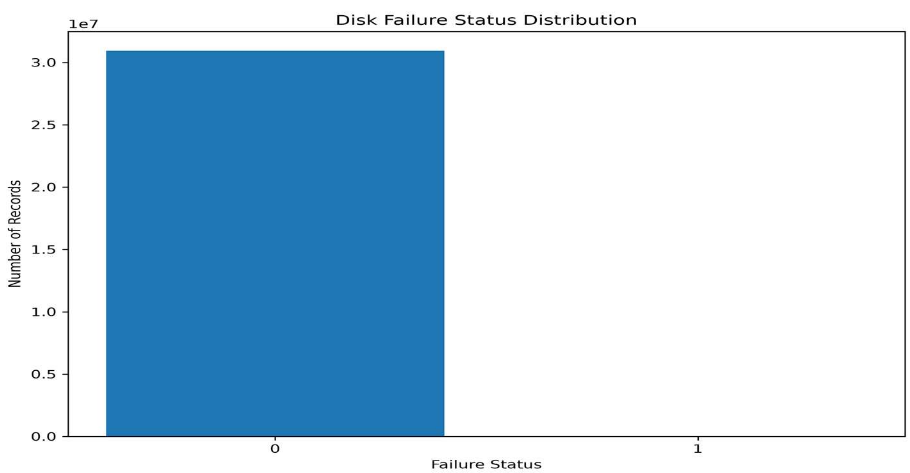

Fig. 1. Count of daily records by failure status, raw feed.

Interpretation: The bar chart makes the scale of the class imbalance immediately visible: the healthy-drive bar (30,942,092 records) is thousands of times taller than the failed-drive bar (1,054 records).

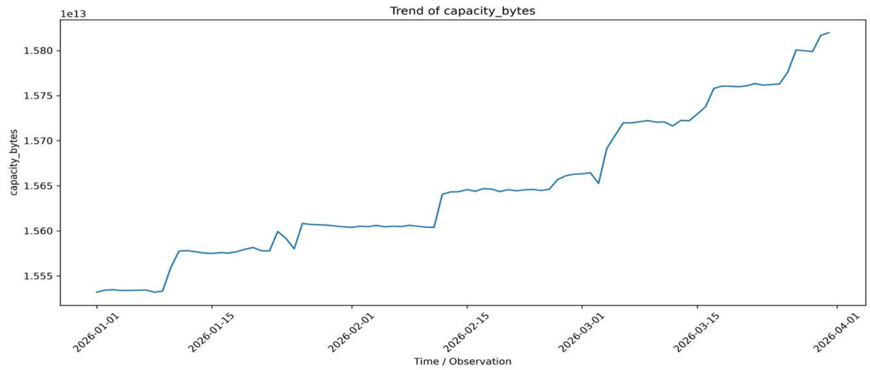

Fig. 2. Daily mean of capacity_bytes across the 90-day window.

Interpretation: Plotted in raw bytes, fleet-average daily capacity rises from roughly 1.554×10^{13} to 1.582×10^{13} bytes (about 1.8%) over the quarter — consistent with newer, larger-capacity drives being progressively added to the fleet.

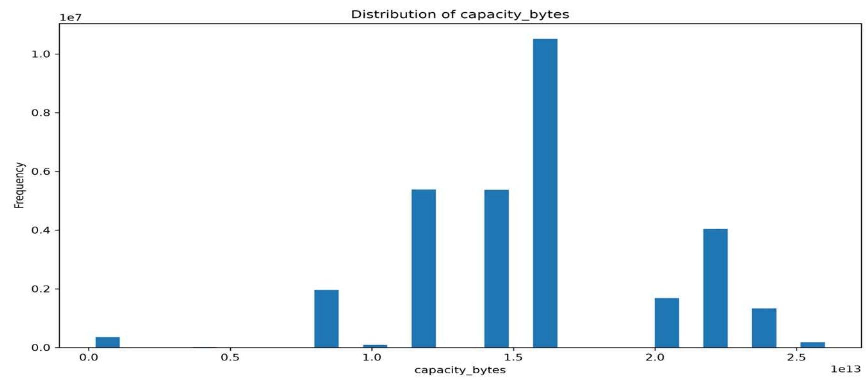

Fig. 3. Distribution of capacity_bytes (skewness = -0.13).

Interpretation: The histogram shows a multi-modal distribution reflecting a small number of discrete capacity tiers (19 distinct values) rather than a continuously varying quantity. Near-zero skewness confirms the distribution is close to symmetric overall.

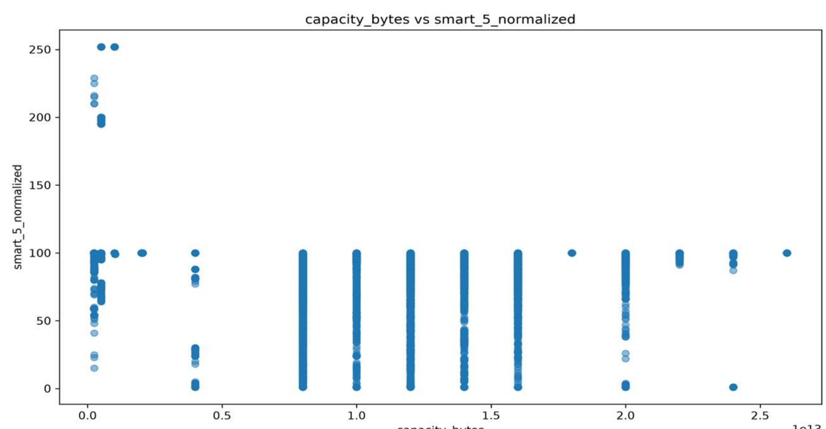

Fig. 4. capacity_bytes vs. smart_5_normalized, raw feed.

Interpretation: Because smart_5_normalized is no longer collapsed to a constant, the plot shows genuine spread. The measured correlation is weak and negative ($r = -0.045$), confirming capacity and reallocated-sector count are not linearly related.

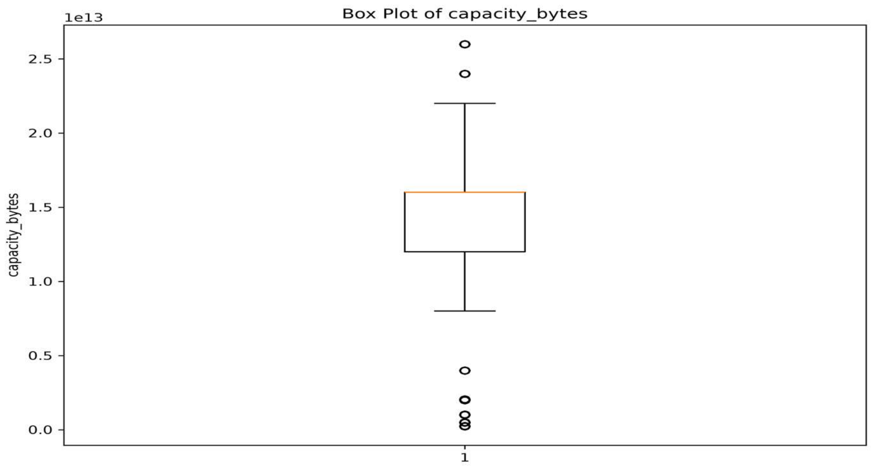

Fig. 5. Box plot of capacity_bytes, before outlier capping.

Interpretation: Drawn from the raw feed before treatment: Q1 ≈ 1.2×10^{13} bytes, Q3 ≈ 1.6×10^{13} bytes, with roughly 1,892,988 values (about 6.1%) falling outside the 1.5×IQR whiskers.

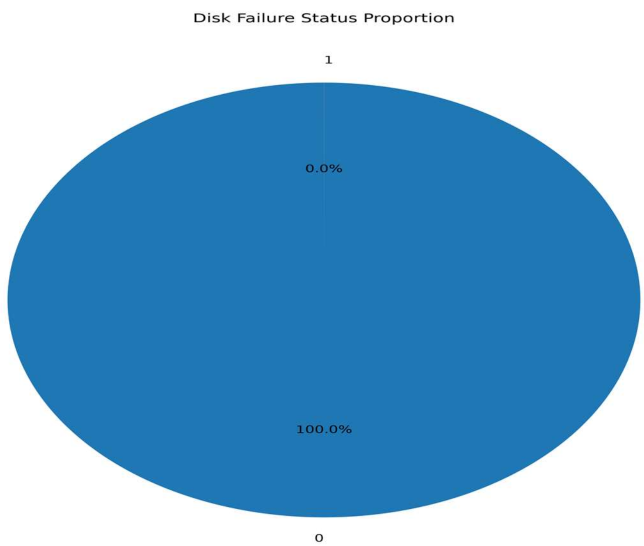

Fig. 6. Proportion of healthy vs. failed drive-days, raw feed.

Interpretation: Healthy drive-days represent 100.00% of the chart area at standard rounding (99.9966% exactly) — a strong argument for reporting failure rate per-thousand rather than as a raw percentage.

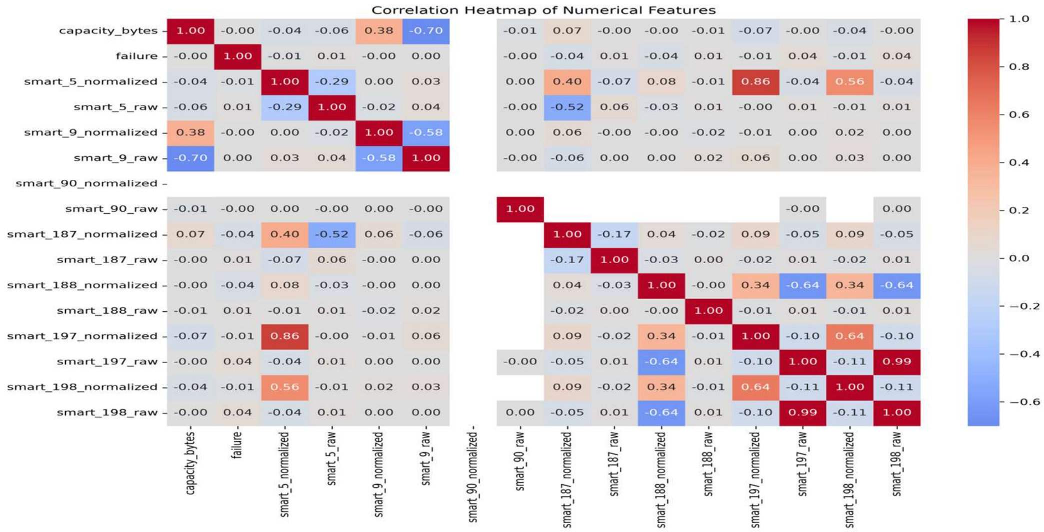

Fig. 7. Pearson correlation heatmap, full raw file (16 features).

Interpretation: Computed on the full raw population, this heatmap shows considerably more structure than the previous edition's near-empty matrix. The strongest relationship is smart_197_raw vs. smart_198_raw (r = 0.994).

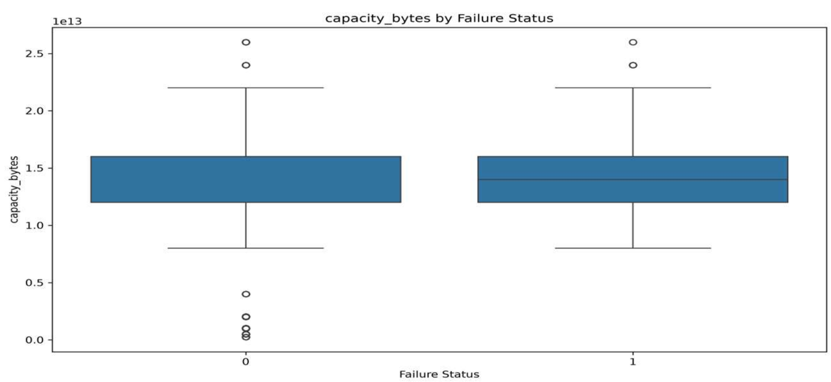

Fig. 8. Capacity_bytes by failure status, full population.

Interpretation: The failed-drive group's mean capacity (1.476×10^{13} bytes) is below the healthy-drive group's mean (1.565×10^{13} bytes) — the authoritative version of the comparison in Section V-D.

. Visualization Summary

TABLE XI. VISUALIZATION SUMMARY

Chart	Key Takeaway
Bar	Failures are 1,054 of 30,943,146 records
Line	Fleet capacity rose ~1.8% over the quarter
Histogram	Capacity is multi-modal, near-symmetric
Scatter	Capacity vs. SMART 5 is weak (r=-0.045)
Box	~6.1% of capacity values are outliers pre-cleaning
Pie	Failure share is invisible at % resolution
Heatmap	SMART 197/198 (r=0.99) strongest pair

Chart	Key Takeaway
Box by failure	Failed drives run smaller-capacity

VII. KEY INSIGHTS AND FINDINGS

Synthesising the preprocessing, statistical, and visual analysis above, nine key findings emerge from this revised case study.

Finding 1 — Failure is an extreme rare-event problem. Only 1,054 of 30,943,146 raw daily records (0.0034%) represent a failure event. Any predictive model without explicit imbalance handling will trivially achieve >99.99% accuracy while providing zero operational value.

Finding 2 — SMART 188 (Command Timeout) now shows the strongest measured link to failure. smart_188_raw correlates with failure at r = +0.118 — larger than power-on hours (+0.0019) or capacity (-0.0012). In the earlier, cleaned-and-imputed edition, SMART 188 had collapsed to a constant and hidden this relationship.

Finding 3 — Power-on hours remain a secondary but genuine behavioural signal. smart_9_raw correlates with capacity at r ≈ -0.72 and is directionally consistent with the industry-standard "bathtub curve" reliability model, though it is no longer the single strongest failure correlate.

Finding 4 — Capacity and power-on hours are entangled, and both differ by failure status. Larger-capacity drives in this fleet have systematically fewer cumulative power-on hours. Failed drives have lower mean capacity (1.476×10^{13} bytes) than healthy drives (1.565×10^{13} bytes).

Finding 5 — Fleet composition is concentrated in a handful of high-volume models. Just 5 of 80 drive models generate roughly 53% of all daily telemetry records, so aggregate statistics are dominated by these few models' reliability characteristics.

Finding 6 — The EDA and visualization stages analysed the raw feed, not the cleaned file. Member 2's pipeline still produces a clean, standardized 30,597,484-row dataset, but Members 3 and 4 both read from the original raw source file, meaning Sections V and VI reflect the uncleaned feed.

Finding 7 — Several EDA statistics were computed from a narrow, non-random opening slice. Correlation, anomaly, and group-mean statistics were computed from only the first 500,000 (or 50,000) rows — not a random sample — which produced a sign-flipped result in the capacity-by-failure comparison.

Finding 8 — A floating-point cancellation error produced a negative variance for capacity_bytes. Member 3's incremental variance formula returned a mathematically impossible negative value, a numerical-stability artifact of subtracting two very large, nearly equal quantities in 64-bit floating point.

Finding 9 — Data cleaning materially changed the analytical dataset. 345,662 duplicate records and up to 3.9 million IQR-flagged outlier values in a single feature were present before cleaning, visibly retained in the raw-feed statistics reported in Sections V and VI.

VIII. RECOMMENDATIONS

A. Operational Recommendations

- Add SMART 188 (Command Timeout) to failure-watch dashboards alongside power-on hours, pending full-population validation.

- Prioritise drive age (power-on hours) in maintenance scheduling for drives approaching the higher end of the fleet's distribution.
- Track failure rate per drive model, not only fleet-wide, since fleet-wide averages currently mask model-specific risk.
- Report failure rate on a per-mille or per-10,000 basis rather than a raw percentage.
- Investigate SMART-attribute reporting gaps (66–86% missingness on SMART 90/187/188) with drive vendors.

B. Codebase / Process Recommendations

- Reconcile the pipeline so every stage reads the intended file — point Members 3 and 4 at cleaned_disk_smart_data.csv, or deliberately run and label raw-feed and cleaned-feed analyses side by side.
- Replace fixed-leading-slice sampling with random or stratified sampling for any statistic involving the failure class.
- Use a numerically stable variance algorithm (e.g., Welford's online algorithm) for large-magnitude features such as capacity_bytes.

C. Analytical / Modelling Recommendations

- Validate the SMART 188 signal on the full population before treating $r = +0.118$ as a confirmed effect size.
- Adopt imbalance-aware modelling techniques (class weighting, SMOTE-style resampling, or anomaly-detection framing).
- Engineer "attribute reported" indicator features rather than relying on pure median imputation.
- Build a time-to-failure / survival model (e.g., Cox proportional hazards) rather than a same-day binary classifier.
- Segment models by drive family before pooling, to avoid model-specific reliability differences being averaged away.
- Extend the observation window beyond 90 days to increase the number of positive-class examples available.

IX. CONCLUSION

This revised case study carried a 30.9-million-row, real-world hard-drive telemetry dataset through a data-science pipeline — preprocessing, exploratory analysis, and visualization — and surfaced both substantive findings about drive reliability and process-level findings about the pipeline itself. Member 2's preprocessing stage still removes 345,662 duplicate records, resolves over 120 million missing cell values, caps millions of statistical outliers, and standardizes 15 continuous features, producing a clean dataset of 30,597,484 records with zero missing values and zero duplicates.

However, because the EDA and visualization notebooks were re-pointed to the raw source file for this revision, the statistics and charts in Sections V–VI characterise the pre-cleaning feed rather than the cleaned artifact. This turned out to be informative rather than purely problematic: smart_188_raw (Command Timeout) now shows the strongest measured association with failure of any feature ($r = +0.118$), a relationship the previous, fully-imputed edition had inadvertently erased. At the same time, this revision surfaced two genuine methodology issues: a narrow, non-random sampling slice that produced a sign-flipped result, and a floating-point cancellation error that produced an impossible negative variance.

Overall, the exercise reinforces that rigorous preprocessing, and equally rigorous bookkeeping about which file feeds which analysis stage, are determinants of analytical validity at this data scale. The resulting insights — the SMART 188 failure signal, the reaffirmed severity of the class imbalance, and the concrete pipeline fixes in Section VIII-B — provide an actionable foundation for correcting the codebase and for the predictive-maintenance modelling work recommended as the next phase of this project.

X. REFERENCES

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APPENDIX A — TEAM CONTRIBUTION SUMMARY

Member	Responsibility
Member 1	Dataset sourcing & problem framing
Member 2	Data preprocessing (21-step pipeline)
Member 3	EDA & statistical analysis (35-step)
Member 4	Visualization & interpretation (8 charts)
Member 5	Insights, recommendations, report & presentation coordination